MIL-B-16907F(SHIPS)
1 April 1963
SUPERSEDING
MIL-B-16907E(SHIPS)
19 July 1960

MILITARY SPECIFICATION

# BOILERS, STEAM, HIGH PRESSURE, AUXILIARY FIRE TUBE TYPE, NAVAL SHIPBOARD

## 1. SCOPE

1.1 Scope. - This specification covers oil fired horizontal straight fire tube type boilers of 2 pass design.

1.2 Classification. - The boilers shall be of the following classes and sizes, as specified (see 6.1):

Class I - Carbon steel construction.
Class II - Nonmagnetic construction.

| Size | Capacity | |
|---|---|---|
| | Rated (see 3.8.2.3) | Overload (see 3.8.2.4) |
| 1 | 500 | ---- |
| 2 | 800 | ---- |
| 3 | 1200 | 1440 |
| 4 | 1500 | 1800 |
| 5 | 2500 | 3000 |
| 6 | 4000 | 4800 |
| 7 | 5000 | 6000 |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of
MIL-S-901 - Shock Tests H. I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use)
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use)
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use
MIL-R-2033 - Relays for Naval Shipboard Electrical Service
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-C-2174 - Controllers, Direct Current, Naval Shipboard Use
MIL-C-2194 - Cable Power, Electrical Reduced Diameter Type, Naval Shipboard
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard
MIL-C-5015 - Connectors, Electric, "AN" Type
MIL-J-5624 - Jet Fuel, Grades JP-4 and JP-5
MIL-P-14276 - Paint, Heat Resisting, Silicone, Aluminum
MIL-P-15024 - Plates, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-M-15071 - Manuals, Equipment and Systems
MIL-P-15137 - Provisioning, Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use)
MIL-I-16165 - Interference Shielding, Engine, Electrical Systems
MIL-P-16298 - Preservation, Packaging, Packing and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts
MIL-W-16878 - Wire Electrical (Insulated, High Temperature)
MIL-F-16884 - Fuel Oil, Diesel, Marine
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits: 14 Kilocycles to 1000 Megacycles
MIL-M-17059 - Motors, Alternating-Current, Fractional HP (Shipboard Use)
MIL-M-17060 - Motors, Alternating-Current, Integral Horsepower (Shipboard Use)



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MILITARY (Cont'd)

MIL-I-17166 - Iron Castings; Nodular Graphitic (Ductile Iron) (For Shipboard Applications)

MIL-P-17286 - Propulsion and Auxiliary, Steam Turbines and Gears (Including Repair Parts Tools, Accessories and Instruments): Packaging of

MIL-M-17413 - Motors, Direct-Current, Integral-HP, Naval Shipboard Use

MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts: Preparation for Delivery of

MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use)

MIL-P-17945 - Pumps, Centrifugal, Peripheral Type, Miscellaneous Naval Shipboard Service

MIL-F-19004 - Fans, Centrifugal, Fixed and Portable, Ventilation, Naval Shipboard

STANDARDS

MILITARY

MIL-STD-22 -- Welded-Joints Designs

MIL-STD-130 - Identification Markings of U.S. Military Property

MIL-STD-167 - Mechanical Vibrations Of Shipboard Equipment

MIL-STD-278 - Welding and Allied Processes For Machinery For Ships Of The United States Navy

DRAWINGS

BUREAU OF SHIPS

810 - 1385850 - Gage Piping Schedule For All Services

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue, at 33rd St., New York 16, N.Y.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler Construction Code

SA-285 - Carbon Steel Plate of Flange and Firebox Quality (2 inches and Under)

(Application for copies should be addressed to the American Society of Mechanical Engineers, 20 West Thirty-Ninth St., New York, N.Y.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction. - Prior to beginning production, one boiler shall be submitted for preproduction inspection specified in 4.2 (see 6.3).

3.2 Electrical characteristics. - The boiler and auxiliaries shall be designed to operate on the electric power supply of the rating specified (see 6.1).

3.3 Weight and physical dimensions. - The boilers, class I and II, complete with insulation and lagging, all auxiliaries, accessories and controls (with exception of motor controllers) shall not exceed the limitations specified in table I.

Table I - Physical limitations

| Size | Capacity | | Length | Width | Height | Dry Weight AC units | Dry Weight DC units |
|---|---|---|---|---|---|---|---|
| | rated (lbs. steam/hr.) | overhead (lbs. steam/hr.) | Feet - inches | Feet - inches | Feet - inches | (lbs.) | (lbs.) |
| 1 | 500 | | 5-0 | 3-3 | 4-6 | 1800 | 2000 |
| 2 | 800 | | 6-0 | 3-6 | 4-9 | 2700 | 3000 |
| 3 | 1200 | 1440 | 6-6 | 3-9 | 5-0 | 3000 | 3400 |
| 4 | 1500 | 1800 | 7-0 | 4-3 | 6-0 | 4000 | 4500 |
| 5 | 2500 | 3000 | 8-0 | 4-9 | 6-9 | 6500 | 7000 |
| 6 | 4000 | 4800 | 9-3 | 5-6 | 7-3 | 8500 | 9000 |
| 7 | 5000 | 6000 | 9-6 | 6-0 | 8-0 | 10,000 | 10,500 |



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3.4 Shock resistance. - All support members and hold-down bolts shall be designed to withstand the shock loadings set forth on figure 1. Shock stresses shall be considered for each direction separately and shall not be combined. Bolts designed to be stressed in shear shall be installed in holes of the following sizes:

| Nominal bolt diameter | Diameter of hole |
|---|---|
| Up to and including 3/4 inch | Nominal bolt diameter plus 1/32 inch |
| Larger than 3/4 | Nominal bolt diameter plus 1/16 inch |

3.4.1 Notes 1, 2 and 3 of figure 1, in their entirety, are applicable in calculating shock forces and in design of all members. Auxiliaries and other items of equipment rigidly supported shall not be attached to two structures which can deflect relative to each other under shock loading.

3.5 Material. - Cast iron shall not be used in any part of the boiler or in any part of the boiler auxiliaries. Nodular iron conforming to MIL-I-17166 may be used in such parts of the boiler auxiliaries where no other substitute for cast iron is practicable provided approval of the specific application is obtained from the bureau or agency concerned.

3.5.1 Class I - carbon steel construction. - Except as otherwise specified herein, material used in construction of class I boilers shall conform to the American Society of Mechanical Engineers (ASME) Boiler Construction Code. Material for shell and tube sheets shall be equal to or better than ASME material SA-285, grade C, firebox quality, with respect to maximum allowable stress taken at 500° F. and weldability.

3.5.2 Class II - nonmagnetic construction. - Boilers specified herein are to be installed on ships whose magnetic signature must be extremely low. Therefore great importance is attached to use of nonmagnetic materials wherever possible without involving extensive development or affecting the design characteristics.

3.5.2.1 All components used in the boiler assembly, other than the boiler shell assembly, shall be manufactured from nonmagnetic materials. Except as otherwise specified herein, material used in the construction of equipment shall be considered nonmagnetic if it has a permeability of less than 1.08 before fabrication and not greater than 2.0 after fabrication. Material used shall be fully identified on preliminary drawings (see 3.13) and on all drawings submitted by the contractor for approval after award of the contract or order (see 3.12). Unless otherwise specified herein, the Boiler Construction Code shall apply.

3.5.2.2 Boiler shells (inner, outer, tube sheets, tube assembly, supports and so forth), shall be carbon steel as specified in 3.5.1.

3.5.2.3 Electrical equipment. - Electrical equipment for class II boilers shall be as follows:

(a) All items of electrical equipment shall be fabricated of nonmagnetic material, except for the following parts:

(1) Parts required in the magnetic circuit of the particular item of equipment.
(2) Structural parts of motors, rotary converters, contactors, relays, thermostats, and pressure switches which would entail extensive modification and redesign to permit substitution of nonmagnetic material.

(b) Electric controllers, solenoids, and switches shall have enclosures and panels of suitable nonmagnetic metals.

(c) Stray magnetic fields caused by electrical circuits shall be kept to a minimum by making the area of electric current loops as small as possible and arranging adjacent loops so that the fields will be in opposition. Current loops shall enclose no magnetic material unless essential for proper performance of equipment.

(d) In alternating current (a.c.) motors, all connections between stator winding connections shall be made as short as possible. The power leads to the stator winding shall be run through the same opening in the motor frame.

3.6 Radio interference suppression. - The combustion and feedwater control system complete, and the circuit component enclosures, wiring conduit, relays, switches, transformers and similar equipment comprising the complete circuit shall comply with basic requirements for electric power equipments in accordance with MIL-E-917 and as specified for the individual equipment herein. Deviations from these requirements shall be submitted to the bureau or agency concerned for approval. The combustion control system complete with ignition system shall be arranged to minimize radio interference, both radiated and conducted through the use of enclosures, shielded conduits, connectors and line filters. External leads shall be shielded to reduce radio interference by shielded metallic conduit of proper diameter with spherical-seat ball and come fittings in accordance with MIL-I-16165. All electrical connections and wiring shall be waterproof. Measurements and limits specified in



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MIL-I-16910 shall be used as a guide in designing for radio interference suppression (R. I. S.) and all equipment used for same shall be subject to approval by the bureau or agency concerned.

3.7 Method of operation. -

3.7.1 Boilers shall be equipped for electric spark ignition, automatic or manual as applicable.

3.7.2 Boilers shall be equipped for automatic feed water regulation.

3.7.3 Boilers of sizes 1 and 2 shall be equipped for automatic "on" and "off" type of burner operation.

3.7.4 Boilers, sizes 3 through 7, shall be equipped for one of the following methods of burner operation, as specified (see 6.1).

(a) Manual operation. - A unit manually operated through turn-down range and manually restarted following shut-down.
(b) Automatic operation. - A unit which will automatically modulate through turn-down range plus automatic restart feature following shut-down.
(c) Semi-automatic operation. - A unit which will automatically modulate through turn-down range without automatic restart feature. The unit must be manually restarted following shut-down.

3.8 Design. -

3.8.1 The boiler shall be oil fired horizontal straight fire tube type of the 2-pass (gas side) design. The boiler shall be either right or left hand as specified (see 6.1). The hand of the boiler shall be determined by the position of the control panel box when facing the firing end of the boiler.

3.8.2 The boiler shall have the following design and performance characteristics:

| | |
|---|---|
| Pressure, design: | |
| For boiler sizes 1 and 2 | 100 psi. |
| For boiler sizes 3 through 7 | 150 psi. |
| Pressure, working: | |
| For boiler sizes 1 and 2 | 35 psi. |
| For boiler sizes 3 through 7 | 125 psi. |
| Pressure, safety valve popping: | |
| For boiler sizes 1 and 2 | 45 psi. |
| For boiler sizes 3 through 7 | |
| First valve | 145 psi. |
| Second valve if used | 148 psi. |
| Capacity, rated, pounds per hour | As specified (see 1.2, 3.8.2.3 and 6.1) |
| Efficiency at rated capacity | 78 percent minimum (see 3.8.2.2) |
| Exit gas temperature at overload capacity: | |
| For boiler sizes 1 and 2 | 550°F |
| For boiler sizes 3 through 7 | 650°F |
| Fuel oil burned | Oil, fuel, Diesel, in accordance with MIL-F-16884, jet fuel JP-5 in accordance with MIL-J-5624 as specified (see 3.8.2.2 and 6.1) |

3.8.2.1 Working pressure of boilers equipped for automatic burner operation shall be approximately the mean between the automatic burner cut-in and cut-off pressures.

3.8.2.2 Performance of the boiler shall be based on burning the following fuel oil as specified (see 6.1).

(a) Diesel fuel oil:

| | |
|---|---|
| Carbon | 85 percent |
| Hydrogen | 14 percent |
| Sulphur | 1 percent |
| Nitrogen | Nil |
| Oxygen and undetermined | Nil |
| Moisture | Trace |
| B.t.u./pound | 19375 |

(b) JP-5 fuel oil in accordance with MIL-J-5624, heating value of 19,500 BTU per pound.

3.8.2.3 Capacity of the boiler is defined as pounds of saturated steam per hour actual evaporation at the specified working pressure with feed water and air temperature of 100°F.

3.8.2.3.1 The moisture content in steam shall not exceed 1 percent at all evaporating rates, including overload operation, when boiler is steaming with the maximum chloride and total solids concentration in boiler water of 550 and 1175 parts per million, respectively.

3.8.2.4 Boilers, sizes 3 through 7 shall be capable of overload operation of, at least, 120 percent of their rated capacity.

3.8.3 Except as otherwise specified herein, the boiler shall be designed and manufactured in accordance with the ASME Boiler Construction Code.

3.8.4 The boiler design shall permit normal operation with a momentary roll up to 45 degrees to either side, or a permanent list of 15 degrees to either side, or a permanent inclination of 5 degrees fore and aft with the boiler installed with longitudinal axis of the shell assembly in fore and aft direction.



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3.8.5 The boiler shall be equipped with all auxiliaries, accessories, and controls to make a self-contained and complete steam generating unit when connected with electric power supply and fuel oil and feed water supply and return lines to be provided by the shipyard. No economizers, air heaters or superheaters shall be fitted to the boiler.

3.8.5.1 The auxiliaries, accessories and controls shall include the following:

Burner (see 3.8.11)
Automatic ignition and combustion control system for boilers of sizes 1 and 2 and for boilers of sizes 3 through 7 when automatic operation is specified (see 3.7, 3.8.12 and 6.1).
Flame failure safety control (see 3.8.11.3 and 3.8.13).
Fuel oil system (see 3.8.14).
Forced draft blower and air regulating mechanism (see 3.8.15).
Feed water system, including feed water treatment tank (see 3.8.16).
Automatic feed water regulator and low water cut-off control (see 3.8.17 and 3.8.13).
Motors and controllers as required by the driven auxiliaries (see 3.8.13 and 3.8.19).
Rotary inverter to supply alternating current for automatic ignition when d.c. power supply is specified (see 3.2 and 6.1).
Safety valves as required (see 3.8.2 and 3.8.5.3.1)
Steam stop valve.
Feed water stop valve.
Feed water check valve.
Bottom blow off valves.
Surface blow off valves.
Salinometer valve.
Air vent valve.
2-Water gage with valves (minimum visible range-10 inches, reflex (separately mounted (see 3.8.5.3).
Steam pressure gage (panel mounted flush), gage piping to conform to Drawing 810-1385850.
Combustion chamber observation window.
Pressure limit control.

3.8.5.2 The component parts of the boiler assembly shall be arranged with respect to each other to permit easy access for servicing or repair and replacement of parts.

3.8.5.3 The water gages shall be located on the front end of the boiler as close to the vertical center line of the boiler as practicable.

3.8.5.3.1 If the safety valve(s) cannot be reached from the boiler operating level, a simple effective mechanical means for raising the valve(s) shall be furnished such that operating controls are at, or adjacent to, firing front of the boiler.

3.8.5.4 Boiler and all auxiliaries shall be adequately supported on a base. Means shall be provided for connecting the base to a ship's deck or foundation.

3.8.5.4.1 The base, base assembly (the main supporting frame, including leg support members, gussets and all reinforcing members) and base assemblies for auxiliaries shall be designed to withstand shock loadings set forth on figure 1 (see 3.5).

3.8.5.5 Suitable provisions shall be made for connecting an air supply duct from the weather to forced draft blower and for connecting a smoke pipe to the boiler smoke outlet.

3.8.6 The boiler shall be equipped with necessary brickwork, insulation and lagging consisting of a metal jacket over the boiler shell insulation.

3.8.6.1 All exterior surfaces of boiler shell assembly which are subject to corrosion shall be given 2 coats of silicone paint conforming to MIL-P-14276, prior to application of insulation and lagging.

3.8.6.2 All parts of the boiler assembly shall be insulated so that, at overload operation, the average surface temperature of the casing shall not exceed 135°F. (front, rear and each side considered separately), based upon an ambient air temperature of 100°F. and a velocity of 0.5 feet per second taken 1 foot away from the casing. Local hot spots due to connections and structural ties in excess of 145°F. shall be submitted to the bureau or agency concerned for approval.

3.8.6.3 Refractory walls and liners of brick construction in furnace and dryback assembly application are not acceptable. Castable or plastic refractory suitable for the temperature of the environment shall be used. The refractory shall be the best commercial grade suited for the particular application, and shall be reinforced by anchor bolts, strips, or nichrome wire as necessary.

3.8.7 The boiler shell assembly shall be of welded construction. The welding shall be in accordance with MIL-STD-278, class A1 pressure vessels.

3.8.7.1 The tube sheet shall be of the flat flanged construction shown on figure P-21(d) of the ASME Boiler Construction Code.

3.8.7.2 Longitudinal seams in the boiler shell and circumferential joints between the tube sheets



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and the shell cylinder and between the tube sheets and the combustion chamber shall be welded from inside and outside and shall comply with the butt joints B-5, B-14 or B-31 of MIL-STD-22, except for the following:

(a) Joints which cannot be welded on inside shall comply with B-11 of MIL-STD-22.
(b) When location of the combustion chamber in the shell renders flanging of the tube sheet for the butt welded joint between the tube sheets and the combustion chamber impracticable a single "J" tee joint, fillet reinforced weld shall be used. This joint shall be similar to joint number T-42, except that the root gap shall not exceed 1/8 inch.

3.8.7.3 The finished diameters of tube holes in tube sheets shall not exceed the following diameters:

| Size tube<br>Inches | Tube hole diameter<br>Inches |
|---|---|
| 1 | 1.02 |
| 1-1/4 | 1.28 |
| 1-1/2 | 1.53 |
| 1-3/4 | 1.78 |
| 2 | 2.03 |
| 2-1/4 | 2.28 |
| 2-1/2 | 2.53 |

3.8.7.3.1 The edges of the tube holes shall be dressed free of all projecting metal and sharp edges.

3.8.7.4 Tubes shall be connected to both front and rear tube sheets by rolling, heading and welding around the edge of the head. Tubes having extended surface (fin-tubes) or gas side turbulators may be used provided approval of their use in the particular design and arrangement is obtained from the bureau or agency concerned.

3.8.8 The boiler shall be equipped with the following internal fittings:

(a) Dry pipe.
(b) Surface blow pipe.
(c) Feed pipe.
(d) Baffle plates as required.

3.8.9 The boiler shell shall have a manhole, as required by the ASME Boiler Construction Code, and a minimum of 6 handholes located so as to permit easy inspection and access to all parts of the shell assembly for flushing of the watersides.

3.8.10 Except as otherwise specified herein, all boiler auxiliaries, accessories and controls shall be in accordance with best commercial practice.

3.8.11 The burner shall be a mechanical atomizing type, either pressure or air atomizing. The burner shall be capable of smokeless operation over the entire boiler operating range when burning fuel oil of the required type.

3.8.11.1 For boiler sizes 1 and 2 the burner shall be arranged for a constant rate fully automatic operation by means of "on" and "off" cycling in accordance with the sequence specified in 3.8.12.1.

3.8.11.2 For boiler sizes 3 through 7 the burner shall be of a variable capacity wide range type capable of operating at all firing rates corresponding to boiler steaming loads from the overload capacity (see 3.8.12.2) to 20 percent of the overload capacity without change of the sprayer plates or burner nozzle tips.

3.8.11.3 When manual or semi-automatic operation is specified (see 3.7), the electric spark ignition and the flame failure control (see 3.8.13) shall be simultaneously energized by manually operated push button.

3.8.11.4 Pilot burner, if furnished, shall be electric-ignited type. The pilot fuel shall be cut off automatically at the end of the trial-for-ignition period of the main burner. The pilot shall be proved prior to delivery of main burner fuel.

3.8.11.5 Air pressure atomizing burners, if used, shall be equipped with an automatic cut-off device which will positively stop boiler operation in the event of air supply failure or any derangement in air supply system resulting in improper atomization. After such emergency cut-off, the boiler shall be restarted only by manually operating the push button master switch on the control panel.

3.8.11.6 An air compressor when required for an air pressure atomizing burner shall be furnished and installed within the space limitations specified for the boiler assembly complete.

3.8.12 The automatic ignition and combustion control system shall be arranged for the following operation:

3.8.12.1 Boilers sizes 1 and 2. -

3.8.12.1.1 The automatic ignition and combustion control system shall be arranged for the following sequence of operation upon the initial start and all subsequent automatic restarts during "on" and "off" cycling:

(a) A pre-purge period of 15 seconds (without ignition spark).



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(b) At the end of the pre-purge period the ignition circuit and the fuel oil solenoid valve shall be energized simultaneously.
(c) The ignition circuit shall be de-energized 15 seconds after opening of the fuel oil solenoid valve.
(d) The blower shall operate through a post-purge period of at least 30 seconds following shut-off of the fuel oil solenoid valve.

3.8.12.1.2 The automatic ignition and combustion control system shall be arranged to shut off the fuel oil solenoid valve 5 seconds (plus or minus 1 second) after opening in case of ignition failure. Following this shut off of the fuel oil solenoid valve, the blower motor shall continue operation for at least 30 seconds to purge the combustion chamber. After completion of the purging period, a safety shutdown of burner operation shall take place. Upon such safety shutdown the burner shall be restarted only by manually operating the push button master switch on the control panel.

3.8.12.2 Boilers sizes 3 through 7.-

3.8.12.2.1 The automatic ignition and combustion control system shall be designed to automatically modulate the burner without "on" and "off" cycling or change of sprayer plates through its entire firing range when the boiler is steaming at variable loads between the overload capacity and 20 percent of the overload capacity, and to automatically cycle the burner "on" and "off" when the boiler is steaming at loads under 20 percent of the overload capacity.

3.8.12.2.2 The automatic ignition and combustion control system shall be arranged for the following sequence of operation upon the initial start and all subsequent automatic restarts during the "on" and "off" cycling:

(a) A pre-purge period of 45 seconds shall be provided. This period shall begin with the burner in the high fire position. The burner shall remain in the high fire position for 15 seconds and then move to the low fire position for the balance of the pre-purge period.
(b) At the end of the pre-purge period, if burner is in the low fire position, the ignition circuit shall be energized.
(c) The fuel oil solenoid valve shall be energized 55 to 60 seconds from the start of the pre-purge period.
(d) The trial-for-ignition period shall be not more than 5 seconds on pilot (if furnished) or on main burner.
(e) The ignition circuit shall be de-energized at the end of 15 seconds after opening of the fuel oil solenoid valve. The burner, then shall move to the position demanded by the modulating control.
(f) The fuel oil solenoid valve shall be instantly de-energized when operating controls call for the burner shutdown.
(g) Post-purge period of at least 30 seconds upon each shutdown of the fuel oil solenoid valve. During the post-purge period the burner shall move to the high fire position to be ready for the next operating cycle.
(h) The overall duration of the programming cycle shall not exceed 120 seconds.

3.8.12.3 The combustion control system shall shut off the fuel oil solenoid valve in 5 seconds after opening in the case of ignition failure. Following this shut off of the fuel oil solenoid valve, the blower motor shall continue operation for at least 30 seconds to purge the combustion chamber. During this period the burner shall move to the high fire position. After completion of the purging period a safety shutdown of burner operation shall take place. Upon such safety shutdown the burner shall be restarted only by manually operating the reset button on the control panel.

3.8.12.3.1 Timing periods of the automatic ignition and combustion control system shall not vary due to ambient temperature variation between 32°F. and 120°F. or due to supply voltage variation of plus or minus 15 percent of the nominal voltage. In case of a failure of supply voltage, the programming control shall automatically recycle when power is restored, following the prescribed sequence when protective limiting devices have been reset.

3.8.12.4 When semi-automatic operation is specified, the automatic modulating feature shall automatically regulate the burner throughout its entire firing range.

3.8.12.5 When automatic operation is specified, following shut-off due to flame failure, the blower motor shall continue operation for at least 30 seconds. During this period, the burner shall move to high fire position. After completion of the purging period the burner shall shut-down on safety. Following such safety shut-down the burner shall be restarted only by manually operating the reset button. Following shut-down for any reason, provision shall be made to insure that the restart can only be made at the low fire position.

3.8.13 The combustion control circuit complete, including the automatic ignition circuit, shall include the following:

(a) Relays (as necessary) (see 3.8.13.2(d)).
(b) Switches (as necessary) (see 3.8.13.2(e) and (f)).



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(c) Control circuit transformer (see 3.8.13.2(i)).
(d) Ignition transformer (3.8.13.2(j)).
(e) Modulating pressure switch (boilers sizes 3 through 7)
(f) Pressure limit switch (steam)
(g) Time delay relays (as necessary)
(h) Pressure switch (air)
(i) Water level control
(j) Combustion safeguard (see 3.8.13.2(h)).
(k) Solenoid valve (as necessary) (see 3.8.13.2(g)).
(l) Sequencing timer or programmer
(m) Modulating motor (see 3.8.13.2(k)).
(n) Ignition electrodes
(o) Fuses (as necessary)
(p) Indicator lights (if used)

3.8.13.1 The control circuit, except for motor starter, shall be suitable for operation with 220 volts a.c. electric supply. A transformer shall be furnished for the control circuit in accordance with available service current (see 3.8.18). Ambient temperature shall be the prime consideration governing location of the control circuit transformer on the boiler assembly. A rotary inverter shall be provided, as necessary, in accordance with available service current (see 3.8.18).

3.8.13.2 All control components, safety devices and circuit or interconnecting wiring shall be as simple as possible and shall be in accordance with MIL-E-917 and the following:

(a) Mercury tube actuated controls are not acceptable.
(b) All components designed for HI-Shock shall also be capable of withstanding type 1 vibration tests specified in MIL-STD-167 without mechanical damages contact chatter or other maloperation.
(c) Electrical wire and cable shall conform to MIL-C-915, MIL-C-2194, MIL-C-5015 and MIL-W-16878
(d) Relays: All relays, except time delay relays, shall be high impact shockproof units in accordance with MIL-R-2033 and the following:
Type - As required
Tripping characteristics - As required
Voltage - As required.
Duty - Continuous or intermittent as required
Ambient temperature - 50°C (122°F.)
(e) Switches: Switches and pushbuttons shall be HI shock design and shall comply with requirements of MIL-C-2212 and the following:
Voltage rating - As required.
Location - Local (mounted on operating panel).
Type - Pushbutton or rotary
Ambient temperature - 50°C., (122°F.)

NOTE: Limit switches will, of necessity, have a remote location mounted near the parent machine. Their enclosure shall be dripproof.

(f) Pressure switches: All pressure switches except the modulating pressure control switch, shall be HI-shock design and shall comply with requirements of C-2212 and the following:
Voltage rating - As required
Location - Remote
Type - Pressure actuated
Ambient temperature - As required (50°C(122°F) minimum).
Enclosure - Dripproof
(g) Solenoid valves: Solenoid valves shall be capable of withstanding HI-shock tests in accordance with type C requirements of MIL-S-901. The solenoid coil shall be designed for continuous duty and the following:
Voltage rating - As required
Ambient temperature - 50°C. (122°F.)
Temperature rise - Shall not exceed values specified in MIL-C-2212
Enclosure - Spray tight or better
Terminal block - As required (fitted in the solenoid enclosure)
Locations - Remote
(h) Combustion safeguard (flame failure safety control): The combustion safeguard may be photo-electric or photo-conductive. The control shall monitor the pilot and the main flame under all firing conditions and de-energize the fuel oil solenoid valve in not more than 2-seconds in event of flame failure. The photo-conductive type shall be designed to discriminate between flame and hot refractory or other source of infrared. The control assembly and the sensing cell or tye shall be tested for HI-shock in accordance with MIL-S-901 and for vibration in accordance with MIL-STD-167.
(i) Control circuit transformer: The transformer shall have winding voltages, frequency and output ratings as required. The transformer shall conform to commercial Marine standards and the following:
Volt ampere rating - As required.
Primary voltage - As specified (see 6.1).
Secondary voltage - 230 V single phase (a.c.)



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Enclosure - Spray tight or better.
Voltage regulation - As required.
Duty - Continuous.
Terminal posts - As required.

(j) Ignition transformer: The ignition transformer shall conform to commercial Marine standards and the following:
Volt ampere rating - As required.
Primary voltage - 230 volts, single phase.
Secondary voltage - 10,000 volts (plus or minus 250 volts).
Enclosure - Spray tight or better for remote mounted units.
Mounting location - Local or remote as required.
Ambient temperature - 50°C. (122°F.).
R.I.S. - As required (see 3.6.).
Duty - Intermittent.

(k) Modulating motor: The modulating motor shall conform to commercial Marine standards and the following:
Timing - 15 or 30 seconds as required.
Stroke travel - As required.
Load torque (ft.-lbs.) - 150 percent of calculated load.
Voltage - As required by circuit design
Location - Remote.
Enclosure - Dripproof.

(l) Rotary inverter: The rotary inverter used to supply AC power required for the boiler control and ignition circuit shall be in accordance with MIL-M-17556 and have the following classification requirements:
Size KW - As required.
Voltage output - 220 volts.
Phase - Single.
Service - Navy A.
Cycle - 60 cycle.
Prime mover - D.C. motor.
Type - Revolving armature.
Enclosure - Dripproof protected.
Speed - Constant.
Duty - Continuous.
Bearings - Ball
Ambient temperature - 50°C (122°F)
Insulation - A, B or H.
R.I.S. - As required.

(m) Low-water cut-off control: The low-water cut-off control shall be one of the following types:
(1) Float type
(2) Electrode - Probe type

(1) Float type: The float type shall be designed for flange mounting on boiler drum or shell. The float cage or surge chamber shall be removable through companion opening in drum or shell. The switch mechanism shall be magnetically operated. The float control shall be a unit which has passed HI-shock and environmental vibration requirements in accordance with MIL-S-901 and MIL-STD-167 respectively.

(2) Electrode probe type: The electrode probe type shall be designed for flange mounting on boiler drum or shell. The probes shall be located and arranged to form an isolated circuit, ungrounded. Each probe shall be mounted independently. Probes, complete with cage or surge chamber shall be removable through companion opening in drum or shell. The electrodes shall be insulated with teflon. A ground indicating circuit shall be wired into the cut-off branch of the control circuit. Arrangement shall be similar to figure 2 and switches and indicators shall be located on the control operating panel:

(n) Motor controllers: Electric controllers (motor starters) shall be in accordance with the following:
(1) They shall be mounted remote to the boiler control circuit enclosure.
(2) MIL-C-2174 for D.C. controllers.
(3) MIL-C-2212 for A.C. controllers.
(4) The controllers shall conform to the following classification requirements:
Service - Navy A (Hi-shock)
Master switch-Spraytight 3 element push button, marked "Start","Stop" and "Emergency Run".
Ratings - As required.
Enclosure - Dripproof.
Operation - Magnetic.
Type - Resistance for D.C. units and across line for A.C. units
Function - Motor starting.
Duty - Continuous.
Protection - Overload protection with emergency run feature and low voltage protection.
Performance - Automatic or semi-automatic as required.
Ambient temperature - 50°C.

(o) Thermoplastic material - Shall not be used.

(p) Console or control enclosure: The enclosure shall be designed in accordance with class I, type dripproof requirements of MIL-E-2036.



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3.8.13.2.1 Following shutdown for any reason, provision shall be made to insure that manual restart can only be made at low fire position. After safety shutdown of the burner, it shall be restarted by manually operating a reset button. Following shutdown for any reason, provision shall be made to insure that manual restart can only be made at low fire position.

3.8.13.3 The controls shall be mounted in a dripproof enclosure. The enclosure may be a console type or mounted on an operating panel consolidating all indicators, lights and pushbuttons thereon. The service of each pushbutton shall be marked. Particular care shall be taken in design and arrangement of all control equipment to insure safety of operating personnel from danger of electric shock. Electric motor controllers (starters) shall be arranged for remote mounting. The main control circuit transformer shall be located outside the control circuit enclosure.

3.8.14 The fuel oil system shall include fuel oil strainer, fuel oil, pump, fuel oil filter (with non-disposable element), relief valve, solenoid oil valve, metering oil valve, as required, supply and return line pressure gages, and interconnecting valves and fittings and pilot burner.

3.8.14.1 Fuel oil strainers and filters shall be either the duplex type or the simplex - self cleaning type so that no interruption in flow can occur when cleaning.

3.8.14.2 The fuel oil pump shall be of the self-priming type and shall have a minimum suction lift of 12 feet. The pump shall be capable of supplying all requisite oil and shall have characteristics suitable for operation with the burner furnished. The pump shall have a direct coupled transmission and a separate motor drive (motor shall have no other loading).

3.8.14.3 When automatic or semi-automatic operation is specified (see 3.8.13.2(n)) the system shall be arranged to permit manual regulation throughout the entire firing range of the burner, in addition to automatic regulation.

3.8.15 The forced draft blower shall be designed to operate with a negative pressure of 3/10 inch of water at the suction end, imposed by resistance in the air supply duct between the blower air intake and the weather, and shall be capable of supplying all requisite air to the boiler at a pressure at least 0.5 and 1.0 inch of water for boilers, sizes 1 and 2 and boilers, sizes 3 through 7, respectively, in excess of the maximum pressure required to overcome all resistances of the burner, and the boiler between the air intake of the air regulating control and the exhaust gas outlet of the boiler. The blower wheel (fan) shall be in satisfactory static and dynamic balance. The interior of the blower casing shall be smooth and free from obstructions, such as bolt heads, nuts and pipe fittings. The following construction features shall apply:

(a) The wheel shall be keyed on the shaft.
(b) Provision shall be made for 2-set screws in the hub located 90 degrees apart.
(c) One set screw shall be located over the key.
(d) One set screw shall be recessed in the shaft for positioning axially.
(e) Frictionless type bearings shall be employed and the shaft shall be provided with a shoulder or collar to prevent axial thrust.
(f) The blower shall have a direct coupled drive. The maximum usable pressure of the fan shall be determined by the total pressure-volume curve in accordance with the procedures specified in MIL-F-19004.

3.8.15.1 An air regulating control shall be provided for controlling air supply to the burner in correct proportion to its firing rates. The control shall be interlocked with the metering oil valve in the fuel oil system. The linkage shall be designed to provide simultaneous manual regulation of the fuel oil and air supplies to the burner when boiler is steaming at variable loads. Means shall be provided in the linkage for manual adjustment of fuel oil-air ratio as necessary for smokeless operation of the burner throughout its entire firing range.

3.8.16 The feed water system shall include a water strainer, feed pump, relief valve, pressure gage, feed treatment tank, feedwater stop and check valves and necessary connecting piping, valves, and fittings.

3.8.16.1 The boiler feed pump shall be of a centrifugal or a turbine type capable of feeding requisite water at 190°F. to the boiler steaming at maximum capacity with a pressure of 150 p.s.i.g. The pump shall be self priming. Pump discharge head shall have a practical margin when injecting water into the boiler through a feed line containing a feed water regulator control valve, and stop and check valves. Design and construction shall be in accordance with MIL-P-17945. The pump shall have a direct coupled drive. The motor drive shall hav no other loading.

3.8.16.2 The feed treatment tank shall be of 1 gallon or more capacity as approved by the bureau or agency concerned and shall be installed in a by-pass branch in the feed water system located between the feed pump relief valve and the feed water stop and check valves. The tank shall be valved so



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that feed water can be passed through the main feed water line or by-passed through the branch containing the tank. The tank shall be equipped with a funnel for adding chemicals, a vent valve, and a drain valve.

3.8.17 The automatic feed water regulator and low water cut-off control shall be of a float type or of an electrode type designed for flange mounting on the top of the boiler shell at a location equidistant from the tube sheets. The control shall be arranged to function without short cycling when water in the boiler is shifting due to conditions specified in 3.8.4. The control shall operate the feed water pump between the low and the high water level limits and shall stop boiler operation when the lowest permissible water level is reached. After this emergency cut-off the boiler shall be restarted only by manually operating the push button master switch on the control panel. The water level shall hold within plus or minus 4 inches at all steaming rates and during on-off operation.

3.8.17.1 Provision shall be made for manual operation of the feet water pump in case automatic operation is not desired or prevented.

3.8.18 Electric service power available for operation of motors shall be as specified (see 3.2 and 6.1).

3.8.19 Electric motors. - Electric motors for driven auxiliaries shall be in accordance with MIL-M-17556, MIL-M-17413, MIL-M-17059 or MIL-M-17060.

3.8.19.1 The motors shall conform to the following classification requirements:

| | |
|---|---|
| Horsepower | As required. |
| Volts, Number of phases, Frequency | As specified (see 3.8.18) |
| Speed | Constant. |
| Duty | Continuous. |
| Enclosure | Dripproof protected. |
| Service | Navy A., |
| Ambient temperature | 50°C. |
| Insulation | A, B, or H. |
| Type | Induction. (For a.c. motors.) |
| Winding | Squirrel cage. (For a.c. motors.) |
| Winding | Shunt (for d.c. motors). |
| Bearing | Ball. |
| Design | B (for a.c. motors). |
| Radio noise suppression | As required (for d.c. motors). |

3.8.20 Wherever practicable the drive and driven shall be direct coupled with flexible coupling. Belt drives may be employed subject to obtaining specific approval from the bureau or agency concerned. Belt drives when employed shall have the following features:

(a) Matched belts shall be furnished for multiple belt drives.
(b) Belt speeds in excess of 5,500 feet per minute are not acceptable.
(c) Drive and driven sheaves shall be in satisfactory static or dynamic balance in accordance with internally excited vibration, type II requirements of MIL-STD-167.
(d) Axial movement in drive and driven shall be controlled by thrust bearings. Thrust bearings may be of the combined radial and thrust rolling contact type.

3.8.21 Identification plates. - Each boiler shall have an identification plate conforming to type A or B of MIL-P-15024, except that identification plates of phenolic compounds are not acceptable. The identification plates shall contain the following information:

(a) Name and description of article.
(b) Serial number.
(c) Federal stock number.
(d) Name and address of manufacturer.
(e) Contract number.
(f) A square of about 3/4 inch with letters "INSP" at the top for affixing Government inspector stamp.
(g) The words "U.S."
(h) Design pressure, p.s.i.
(i) Operating pressure, p.s.i.
(j) Hydrostatic test pressure, p.s.i.
(k) Safety valve setting, p.s.i.
(l) Electrical characteristics.

3.8.21.1 The stock number will be furnished to the contractor by the bureau or agency concerned on written request.

3.9 Repair parts. - The following repair parts shall be furnished as specified (see 6.1):

(a) Onboard repair parts.
(b) Stock repair parts.

3.9.1 Onboard repair parts. - The set of repair parts for boiler and auxiliaries and the set of repair parts for electrical equipment, shall be as specified herein.

3.9.2 The number of sets of repair parts shall be as specified herein.

3.9.3 The set of repair parts for boiler and auxiliaries shall include the following:

(a) Boiler parts:

| | Percent |
|---|---|
| Manhole gaskets (if required) | 800 |
| Handhole plates (each type and size) | 25 |



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| | | Percent |
|---|---|---|
| (a) | Boiler parts: (Cont'd) | |
| | Handhole gaskets (each type and size) | 800 |
| | Boiler water gage glass sets | 400 |
| | Gaskets - complete sets (required for boiler and burner assembly) | 800 |
| | Drive belt sets (matched sets required | 200 |
| | Safety valve internals - sets (including seats, discs and springs) | 200 |
| (b) | Burner parts - | |
| | Atomizer barrel | 200 |
| | Sprayer and Orifice plates or nozzle tips | 600 |
| | Metering oil valve | 100 |
| | Pilot burner tips | 600 |
| (c) | Blower parts - | |
| | Bearings - set | 100 |
| (d) | Feedwater strainer parts - | |
| | Strainer baskets | 200 |
| (e) | Fuel oil pump parts - | |
| | Pump (complete assembly) | 100 |
| | Pump relief valve | 200 |
| | Pump shaft coupling | 200 |
| | Coupling inserts | 600 |
| | Seal assemblies | 200 |
| (f) | Air pump or compressor (if used) | |
| | Air pump (assembly) | 100 |
| (g) | Feedwater pump parts - | |
| | Shaft (without impeller) | 100 |
| | Impellers (set) | 100 |
| | Pump shaft coupling | 100 |
| | Coupling inserts | 600 |
| | Packing | 200 |
| | Gaskets | 200 |
| (h) | Feedwater regulator and low water cut-off parts - | |
| | Magnetic float type: | |
| | Switch assembly | 100 |
| | Float, rod and magnet | 100 |
| | Electrode probe type: | |
| | Electrode plugs, sets | 300 |
| (i) | Electric and electronic equipment parts - | |
| | Motors, set for each type and size in accordance with MIL-M-17413 or MIL-M-17556, as applicable for D.C. motors, including 1 armature complete with commutator and shaft | 100 |
| | Motor, set for each type and size in accordance with MIL-M-17060 or MIL-M-17059, as applicable (stator coils and rewind material shall not be included) | 100 |
| | Rotary inverter (if furnished) set for each type and size in accordance with MIL-M-17556 | 100 |

| | | Percent |
|---|---|---|
| (i) | Electric and electronic equipment parts - (Cont'd) | |
| | Controller, D.C., set for each type and size in accordance with MIL-C-2174 | 100 |
| | Controller, A.C., set for each type and size in accordance with MIL-C-2212 | 100 |
| | Relays of each type and size, including time delay units | 100 |
| | Ignition transformers | 100 |
| | Ignition electrodes | 400 |
| | Ignition cables and connectors | 400 |
| | Combustion safeguard | 100 |
| | Vacuum tubes of each type and size | 400 |
| | Combustion safeguard sensing element | 400 |
| | Modulating motor | 100 |
| | Oil solenoid valve | 100 |
| | Oil solenoid valve coils | 400 |
| | Mechanical or electric timers or programmers | 100 |

3.9.4 Stock repair parts. - Stock repair parts shall be furnished in accordance with MIL-P-15137.

3.9.5 Marking for identification. - Identification marking of repair parts shall be in accordance with MIL-STD-130.

3.10 Tools. -

3.10.1 The set of tools and wrenches, special, shall be as specified herein. Each tool shall be indelibly marked to indicate the service for which intended.

3.10.2 The number of sets of special tools and wrenches, shall be as specified (see 6.1). Special tools are defined as those tools not listed in the Federal Supply Catalog (This catalog may be consulted in the office of the Government inspector)

3.10.3 The set of tools and wrenches, special, shall include the following:

1-Tube expander and mandrel for each size of tube used.
1-Manhole cover wrench, if required.
1-Handhole cover wrench.
1-Ripping chisel.
1-Oyster knife.
1-Set of tube plugs (blind nipple type for use with expander and mandrels furnished for tubes) consisting of 10 percent of the number of each size of tube installed in the boiler.
1-Set of safety valve gages.



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1-Safety valve seat and disc templates.
1-Set of gasket templates (one for manhole, if required, and one for each size of handhole).
The templates shall be of steel 1/16 inch thick.
2-Brushes, wire-bristle, fire tube cleaning.
1-Brush handle, sectionalized, for fire tube cleaning brush.
1-Manual ignition torch with torch holder.
1-Set of feed water pump tools and wrenches as required.
1-Set of tool boxes (metal) as required.

3.11 List of repair parts and tools and wrenches.- Repair parts list shall be in accordance with MIL-P-15137.

3.12 Drawings.-

3.12.1 Drawings shall be submitted to the Bureau or agency concerned within thirty days of the date of contract. Submittal, review, approval and validating procedure and requirements shall be in accordance with MIL-D-963. The use of manufacturer's standard drawings together with certification data sheets to show design variation is encouraged. Drawings shall contain views, details, data and tabulations as specified herein on drawings entitled as follows:

(a) Assembly and foundation plan
(1) Assembly views (front, rear and side elevations), plan view and foundation of the boiler complete with all auxiliaries, accessories and controls. All components of the assembly shall be identified by piece numbers and listed in the list of material. Overall dimensions, locations and type and size of all terminal connections, and space required for tube renewal shall be shown. The following data shall be tabulated:
Pressure designed, p. s. i.
Pressure operating, p. s. i.
Pressure safety valve setting, p.s.i.
Pressure hydrostatic test, p. s. i.
Heating surface, sq. ft.
Furnace volume, cu. ft.
Evaporating capacity.
Rated
Overload
Oil rate, pounds/hr.
At rated capacity
At overload capacity
Weight dry, pounds
Weight wet, pounds
Electrical characteristics
Exit gas temperature at overload
Efficiency at rated capacity
Design rate of combustion KB/HR per sq. ft. heating surface
Design rate of combustion KB/HR per cu. ft. furnace volume.

(b) Master list of material and tools
(1) List of all items and sub-assemblies giving number of drawing whereon item breakdown for each sub-assembly is shown.
(2) List all tools furnished. The List of material and tools drawing shall be arranged as follows:
Name
Number required
Material
Material specification
Contractor's drawing number
Contractor's part number
Manufacturer's name
Manufacturer's part number
Bureau of Ships drawing number
Unit weight
Remarks

(c) Boiler construction and weld details
(1) Section views of the boiler shell assembly as necessary to show arrangement and details of pressure parts internal fittings and external fittings. Show detail views of all welds. Show methods of connecting tubes to tube sheets. Show stress calculations for the pressure vessel.

(d) Boiler sectional views
(1) Section (side and end) views of the boiler assembly complete with burner front, burner and rear cover. Identify refractory, insulation and lagging; show quantities used and give installation instructions.

(e) Burner assembly and details
(1) Section view of boiler front end with burner installed. Show and identify all constituent parts. Show detail views of each burner part.

(f) Boiler support foundation
(1) Plan view and side and end elevation views of the boiler base complete with boiler shell support members fully dimensioned.

(g) Boiler wiring diagrams
(1) Show wiring diagram and schematic wiring diagram for the boiler auxiliaries, operating controls and safety controls. Identify all symbols. Identify all components in material table. Describe step-by-step the start up, operating controls. Describe function of each safety device, how it is integrated into the system and the



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sequence or change in sequence for each safety shut down.

(h) Sectional assembly views of the following components complete with material table.
  (1) Feedwater regulator and low water cut-off
  (2) Rear cover assembly (dryback)
  (3) Front end plate with insulation.
  (4) Oil pump complete with performance curves.
  (5) Water pump complete with performance curves.
  (6) Burner-blower complete with performance curves.
  (7) Electric motors complete with design characteristics and performance curves.
  (8) Motor controllers showing design characteristics and the internal wiring diagram
  (9) Belt drive assembly - show arrangement and all design data

3.12.2 All work done prior to approval of drawings shall be entirely at the contractor's risk.

3.13 Preliminary drawings. - Each bidder shall submit preliminary drawings and data, as specified (see 6.2), with his bid to adequately describe the equipment he proposes to furnish. If a bidder proposes to furnish equipment which previously has been supplied and for which the bureau or agency concerned has approved drawings on record, reference to the approved drawings will be acceptable in lieu of preliminary drawings.

3.14 Manuals. - Manuals shall be in accordance with MIL-M-15071. Content of the manual shall be arranged in accordance with MIL-M-15071, except that parts identification is not required. In lieu thereof, a "List of Components" comprising major parts of the boiler and burner assemblies and all individual auxiliaries, accessories, and controls furnished with the boiler, shall be included. The "List of Components" shall be arranged as follows:

Column 1 - Assembly piece number.
Column 2 - Manufacturer's part number.
Column 3 - Quantity per boiler.
Column 4 - Name in full (abbreviations shall not be used).
Column 5 - Description containing application or function, material, size, rating, manufacturer's type, or model number, catalog number.
Column 6 - Federal stock number (to be added when available).
Column 7 - Manufacturer's or contractor's drawing num.
Column 8 - Weight.

3.14.1 The list of repair parts and the list of tools and wrenches furnished shall be included in the manual. The arrangement used for these lists shall be the same as specified for the "List of Components".

3.14.2 Reduced size reproductions of the following drawing shall be included in the manual:

(a) Assembly views of the boiler, plan view and foundation.
(b) Sectional views of the boiler.
(c) Arrangement of boiler brickwork.
(d) Arrangement and sectional views of the burner unit.
(e) Assembly and sectional views of all auxiliaries.
(f) Diagrammatic equipment wiring and elementary schematic wiring diagrams.
(g) Master drawings for motors.
(h) Master drawings for controllers.

3.14.3 Manufacturers' standard bulletins applicable to auxiliaries, accessories, and controls furnished shall be included in the technical manual where necessary for complete description of construction, method of installation, operation, adjustment, and maintenance.

3.14.4 The method of approval of technical manual shall be as specified in MIL-M-15071, except that two sample copies of the manual complete, shall be submitted for approval to the bureau or agency concerned.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction. - One boiler shall be submitted to a laboratory satisfactory to the Bureau of Ships for preproduction tests. The boiler shall be tested as specified in 4.2.1.

4.2.1 Tests. -

4.2.1.1 Test agenda. - Prior to testing, a test agenda shall be prepared and submitted to the Bureau of Ships for approval. The agenda shall include all of the tests and requirements specified in 4.2.1.2 and 4.2.1.3 and shall also include the following:



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(a) Diagrammatic sketch of the test layout showing location for each reading to be observed.
(b) Instrumentation complete for each item listed in 4.2.1.3 giving make, model and manufacturer of instrument employed.

4.2.1.2 Preproduction tests shall be conducted in accordance with table II. The tests are required to establish compliance with the requirements of this specification and to ascertain fulfillment of the capacity and efficiency requirements and satisfactoriness of control features and safety devices.

Table II - Tests

| Test | Preproduction unit | Production unit |
|---|---|---|
| Steady steaming at 120 percent capacity | 8-hours consecutively | 1-hour |
| Steady steaming at rated capacity | 24-hours consecutively | 1-hour |
| Steady steaming at 50 percent rated capacity | 1-hour | 1-hour |
| Steady steaming at minimum capacity | 8-hours | 1-hour |
| Steaming at loads modulated from 120 percent capacity to minimum capacity | 8-hours consecutively | ------ |
| Steaming at low loads "on-off" cycling | 300-cycles consecutively | 10-cycles consecutively |
| Safety valve by steam | 1-test | 1-test |
| Combustion safeguard | 10-cycles consecutively | 5-cycles consecutively |
| Low water cut-off | 10-cycles consecutively | 5-cycles consecutively |
| Air-pressure cut-off | 10-cycles consecutively | 5-cycles consecutively |
| Low voltage | 1-cycle | 1-cycle |
| Chemical and physical properties of oil | 1-sample | ------- |

4.2.1.3 During the operational runs specified in 4.2.1.2, the following data shall be obtained and recorded every 15-minutes, except for the steady steaming run on the preproduction unit where data shall be obtained and recorded once every hour. Efficiency of the boiler shall be calculated. Performance curves of the boiler shall be plotted. Data are required as follows:

(a) Oil consumption.
(b) Steam quality.
(c) Blower discharge pressure. 1/
(d) Gas pressure at rear head. 1/
(e) Boiler oil pressure.
(f) Feedwater pump discharge pressure.
(g) Blower air inlet temperature. 1/
(h) Gas temperature at rear head. 1/
(i) Boiler casing temperature. 1/
(j) Motor amperes.
(k) Gas analysis $CO_2$, $O_2$ and CO.
(l) Water level in gage glass.
(m) Water input or steam input.
(n) Burner plenum pressure.
(o) Blower inlet pressure or barometric pressure (as applicable). 1/
(p) Gas pressure at breeching outlet.
(q) Return line oil pressure.
(r) Steam pressure.
(s) Feedwater temperature.
(t) Gas temperature at breeching outlet.
(u) Front and rear casing head temperature.
(v) Line voltage.
(w) Dry and wet weight.
(x) R.p.m. of each drive and driven.
(y) Total electric power consumption.

1/ Not required on the production units.



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4.3 Production inspection. - Production inspection shall be conducted on each boiler and shall consist of the tests specified in table II and the following:

(a) Operational tests as specified in 4.2.1.3
(b) Examined and weighed to determine compliance with the approved drawings.
(c) Hydrostatically tested to 150 percent design pressure.
(d) Examined to determine compliance with section 5 herein.
(e) Where other specifications form a part of this specification, sampling, examination and tests shall be conducted as required in the referenced specification.

## 5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging of the boiler detached auxiliaries and accessories shall be sufficient to afford adequate protection against corrosion, deterioration, physical and mechanical damage during shipment from the supply source to the using activity until early installation and may conform to the suppliers commercial practice when such meets these requirements. All openings shall be sealed to prevent entrance of foreign matter.

5.1.1.2 Packing. - The boiler shall be shipped assembled as a complete unit, except such auxiliaries and accessories that require separate packing for preservation, packaging and security in shipment. Such parts shall be installed by the shipyard. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts and tools. -Onboard repair parts and tools shall be preserved and packaged level A, packed level C and marked levels A and C in accordance with the applicable specifications as follows:

(a) Mechanical parts and tools. - MIL-R-196 and MIL-P-17286.
(b) Electrical parts. - MIL-P-16298 and MIL-E-17555.

Items not specifically covered therein shall be unit protected and packaged in accordance with the guide lines of MIL-P-116. Unless otherwise specified (see 6.1), repair parts and tools shall be individually packaged, except when used in sets.

5.2 Domestic shipment and storage or overseas shipment requirements. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 The boiler and accessories shall be preserved and packaged in accordance with the level A requirements of MIL-B-3180.

5.2.1.2 Repair parts and tools. - Repair parts and tools shall be preserved and packaged level A in accordance with MIL-R-196, MIL-P-17286, MIL-P-16298, MIL-E-17555 as applicable. Items not covered therein shall be unit protected and packaged in accordance with the guide lines of MIL-P-116. Unless otherwise specified (see 6.1) parts shall be individually packaged except when used in sets.

5.2.1.3 Packing. -

5.2.1.3.1 Equipment. -

5.2.1.3.1.1 Level A. - Boilers shall be packed in sheathed crates conforming to MIL-C-104.

5.2.1.3.1.1.1 Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed in overseas type wood cleated plywood, nailed wood or fiber boxes conforming to PPP-B-601, PPP-B-621 or PPP-B-636 (class 2), respectively, at the option of the contractor. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds. Single items having a gross weight exceeding 200 pounds shall be packed in boxes conforming to PPP-B-601 or PPP-B-621, modified in accordance with the applicable box specification. Boxes shall be lined with a sealed caseliner conforming to MIL-L-10547.



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5.2.1.3.1.2 Level B. - Boilers shall be packed in sheathed crates conforming to MIL-C-104. Boilers having a gross weight less than 10,000 pounds may be individually packed in unsheathed crates conforming to MIL-C-3774 with the following additional requirements:

(a) The burner or front end of the boiler shall be completely enclosed in a lumber or plywood housing. The housing shall be anchored to the skid base of the crate and securely strapped to the boiler.

(b) The entire contents of the crate shall be shrouded approximately two-thirds of the way down with waterproof barrier material conforming to UU-P-271. The shrouding shall be securely fastened and arranged in a manner to prevent the formation of water pockets.

5.2.1.3.1.2.1 Detached boiler auxiliaries, accessories and controls which are shipped (knocked down) for installation shall be packed as specified in 5.2.1.3.1.1.1. Boxes shall be of the domestic type or class. Caseliners shall not be required.

5.2.1.3.2 Repair parts and tools. - Repair parts and tools shall be packed level A or B as specified in accordance with the applicable specification specified in 5.2.1.2.

5.2.1.3.3 Anchoring, blocking, bracing and cushioning shall be in accordance with JAN-P-100 and the appendix to MIL-C-104.

5.2.1.4 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.1.4.1 Unpacking instructions shall be provided and shall be sealed in waterproof envelopes which shall be tacked, stapled or otherwise securely attached to the outside of each basic equipment container in the most protected location. The words "UNPACKING INSTRUCTIONS" shall be stenciled on each containers indicating location of such instructions.

5.2.1.5 Manuals. - Manuals shall be packaged and packed for the level of protection specified accordance with MIL-M-15071).

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class, size and capacity of boiler (see 1.2 and 3.8.2).
(c) Electric power supply available for operation of the boiler and auxiliaries (see 3.2, 3.8.5.1, and 3.8.18).
(d) Type of burner operation for boiler sizes 3 through 7 (see 3.7.4 and 3.8.5.1).
(e) Whether the boiler shall be right or left hand (see 3.8.1).
(f) Type of fuel to be burned (see 3.8.2 and 3.8.2.2).
(g) Primary voltage required for control circuit transformer (see 3.8.13.2(i)).
(h) Kind and number of sets of repair parts required (see 3.9).
(i) Number of sets of special tools and wrenches required (see 3.10.2).
(j) Preliminary drawings and data required (see 3.13).
(k) Level of preservation, packaging, packing and marking required (see 5.1 and 5.2).

6.2 Bid data. - The bidders shall furnish with their proposals the following data in triplicate:

(a) Rated capacity of the boilers, pounds steam/hour, normal and maximum, at design conditions (see 3.8.2.3).
(b) Heating surface, square feet.
(c) Furnace volume, cubic feet.
(d) Anticipated performance data at the rated normal and maximum capacities (efficiency, $CO_2$ content and temperature of exit gas, and pounds of oil per hour).
(e) Complete description of the boiler, burner, and auxiliaries.
(f) Complete description of controls and safety devices (manufacturer's bulletins shall be attached).
(g) Weight, dry and wet, of the complete unit.
(h) Itemized list of proposed onboard repair parts.
(i) Itemized list of tools and wrenches, special.
(j) Estimated weights of onboard repair parts and tools and wrenches.
(k) Three sets of blueprints of the following preliminary drawings (see 3.13).
  (1) The outline assembly views (front, sides, and rear) of the boiler complete with auxiliaries, accessories, and controls, giving overall dimensions, locations of all piping connections, and spaces required for tube renewal.
  (2) Sectional elevation and plan views of the boiler and burner assembly showing arrangement of tubes and internal fittings and giving sizes of boiler tubes, thicknesses of shell plate, and identifying refractories and insulation used.



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(1) Satisfactory evidence to show that the bidder has had adequate experience in manufacture of boilers of the type and size offered in the bid. A list of actual marine installations with period of time in service shall be submitted.

6.2.1 That right is reserved to reject bids lacking in full or part the data required in 6.2.

6.3 Preproduction.- Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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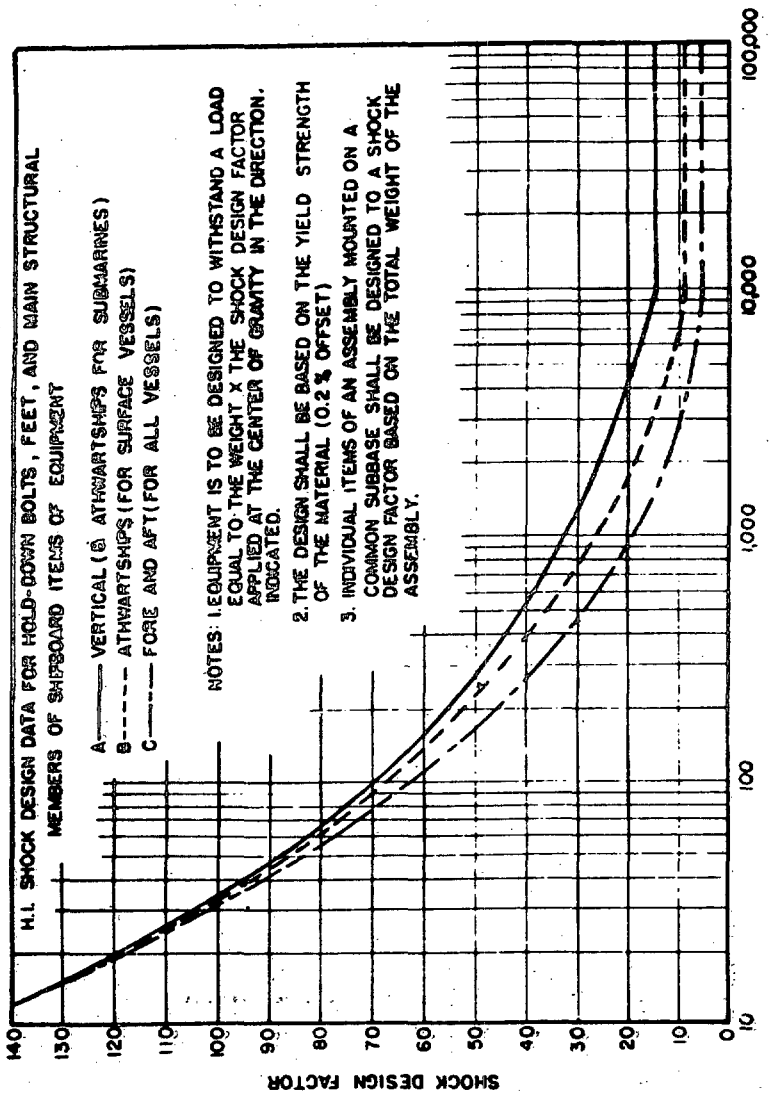
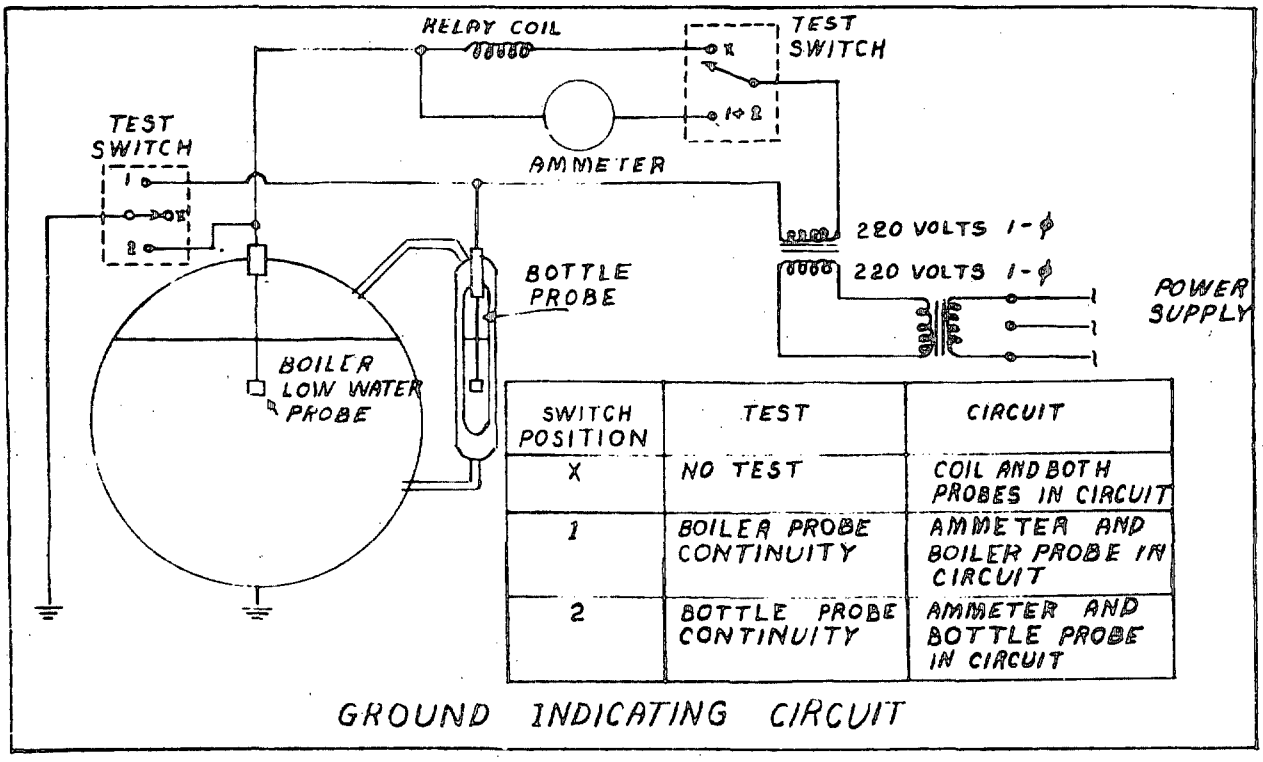



Figure 1 - Shock design data.



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| SWITCH POSITION | TEST | CIRCUIT |
|---|---|---|
| X | NO TEST | COIL AND BOTH PROBES IN CIRCUIT |
| 1 | BOILER PROBE CONTINUITY | AMMETER AND BOILER PROBE IN CIRCUIT |
| 2 | BOTTLE PROBE CONTINUITY | AMMETER AND BOTTLE PROBE IN CIRCUIT |

Figure 2. Ground indication circuit.



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INDEX


| | Paragraph | Page |
|---|---|---|
| Boilers, sizes 1 and 2 | 3.3, 3.7.3, 3.8.5.1, 3.8.11.1, 3.8.12.1 | 2, 4, 5, 6 |
| Boilers, sizes 3 through 7 | 3.3, 3.7.4, 3.8.2.4, 3.8.5.1, 3.8.11.2 3.8.12.2 | 2, 4, 5, 6, 7 |
| Class 1 | 1.2, 3.5.1 | 1, 3 |
| Class 2 | 1.2, 3.5.2 through 3.5.2.3 | 1, 3 |
| Controllers | 3.8.13 | 7 |
| Design | 3.8 | 4 |
| Drawings | 3.12 | 13 |
| Drawings, preliminary | 3.13, 6.2 | 14, 17 |
| Electrical characteristics | 3.2, 3.8.18 | 2, 11 |
| Identification plates | 3.8.21 | 11 |
| Inspection, preproduction | 4.2, 6.3 | 14, 18 |
| Inspection, production | 4.3 | 16 |
| Repair parts, tools and wrenches | 3.9 through 3.11 | 11 through 13 |
| Manuals | 3.14 | 14 |
| Material | 3.5 | 3 |
| Method of operation | 3.7 | 4 |
| Motors | 3.8.13.2, 3.8.19 | 8, 11 |
| Ordering data | 6.1, 6.2 | 17 |
| Preparation for delivery | 5 | 16 |
| Radio interference suppresion | 3.6 | 3 |
| Shock resistance | 3.4, figure 1 | 3, 19 |




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